



US012401207B2

(12) **United States Patent**
Hall et al.

(10) **Patent No.:** **US 12,401,207 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **ENERGY STORAGE SYSTEM AND METHOD
EMPLOYING SECOND-LIFE ELECTRIC
VEHICLE BATTERIES**

(71) Applicant: **B2U Storage Solutions Inc.**, Santa
Monica, CA (US)

(72) Inventors: **Freeman Stoflet Hall**, Santa Monica,
CA (US); **Edward Adolph Becker**,
Champaign, IL (US); **Michael Joseph
Stern**, Westlake Village, CA (US)

(73) Assignee: **B2U Storage Solutions Inc.**, Santa
Monica, CA (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 730 days.

(21) Appl. No.: **17/675,456**

(22) Filed: **Feb. 18, 2022**

(65) **Prior Publication Data**

US 2022/0209546 A1 Jun. 30, 2022

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/382,933,
filed on Jul. 22, 2021, now Pat. No. 11,289,921,
(Continued)

(51) **Int. Cl.**
H02J 7/00 (2006.01)
B60L 50/60 (2019.01)
(Continued)

(52) **U.S. Cl.**
CPC **H02J 7/0024** (2013.01); **B60L 50/66**
(2019.02); **B60L 53/30** (2019.02); **B60L 53/60**
(2019.02);
(Continued)

(58) **Field of Classification Search**
CPC H02J 7/0024
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,547,980 B2 6/2009 Harrison
7,710,073 B2 5/2010 Yamauchi et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 107305972 A 10/2017
JP 2007141464 A 6/2007
(Continued)

OTHER PUBLICATIONS

Notification of Transmittal of the International Search Report (Forms
PCT/ISA/220 and PCT/ISA/210) and the Written Opinion of the
International Searching Authority (Form PCT/ISA/237) issued May
2, 2022, by the International Application Division Korean Intellect-
tual Property Office in corresponding International Application No.
PCT/US2022/011411. (10 pages).

(Continued)

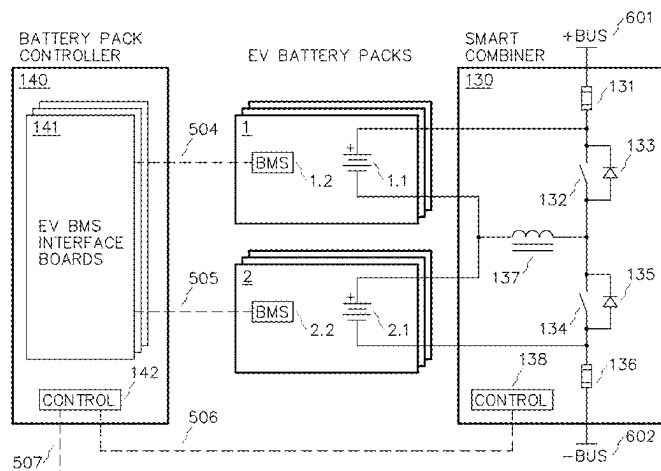
Primary Examiner — Jerry D Robbins

(74) *Attorney, Agent, or Firm* — Panitch Schwarze
Belisario & Nadel LLP

(57) **ABSTRACT**

An energy storage system and method employing second-
life electric vehicle batteries. The system includes a plurality
of electric vehicle battery packs; and a processor configured
to: couple the plurality of electric vehicle battery packs in a
series/parallel arrangement, the series/parallel arrangement
including a plurality of series strings of electric vehicle
battery packs, each of the plurality of series strings of
electric vehicle battery packs includes at least two of the
plurality of electric vehicle battery packs coupled in series,
and the plurality of series strings are connected in parallel;
and wherein the coupling of the plurality of electric vehicle
battery packs includes one or more of connecting electric
vehicle battery packs with lower voltages in series, connect-
ing electric vehicle battery packs with higher voltages in
series, connecting electric vehicle battery packs with major-

(Continued)



ity voltages in series, and connecting electric vehicle battery packs within a programmed voltage connection window in parallel.

20 Claims, 10 Drawing Sheets

Related U.S. Application Data

which is a continuation of application No. 17/118,497, filed on Dec. 10, 2020, now abandoned.

- (51) **Int. Cl.**
B60L 53/30 (2019.01)
B60L 53/60 (2019.01)
B60L 53/80 (2019.01)
- (52) **U.S. Cl.**
 CPC **B60L 53/80** (2019.02); **H02J 7/00032**
 (2020.01); **H02J 7/0014** (2013.01); **H02J**
7/00304 (2020.01); **B60L 2210/10** (2013.01);
B60L 2210/30 (2013.01); **B60L 2210/40**
 (2013.01); **H02J 2207/20** (2020.01)
- (58) **Field of Classification Search**
 USPC 320/109
 See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

8,838,313	B2	9/2014	Bolenbaugh	
9,347,996	B2	5/2016	Uchida et al.	
9,431,832	B2	8/2016	Ichikawa et al.	
10,183,583	B2	1/2019	Narla	
10,411,500	B2	9/2019	Qin et al.	
10,752,128	B1	8/2020	Langton et al.	
10,759,287	B2	9/2020	Vahedi et al.	
10,759,289	B2	9/2020	Gou	
10,763,692	B2	9/2020	Pelletier et al.	
10,784,702	B2	9/2020	Kudo et al.	
10,926,644	B1 *	2/2021	Willson	H02J 7/342
2007/0187957	A1	8/2007	Harrison	
2012/0133337	A1 *	5/2012	Rombouts	G07F 15/006 320/155
2013/0090872	A1 *	4/2013	Kurimoto	B60L 58/13 702/63
2014/0012443	A1	1/2014	Bolenbaugh	
2014/0015532	A1	1/2014	Uchida et al.	
2014/0184142	A1	7/2014	Bito	
2015/0093611	A1 *	4/2015	Obata	G01R 31/392 429/61
2016/0178678	A1	6/2016	Pelletier et al.	
2016/0185246	A1	6/2016	Paul	
2016/0240898	A1 *	8/2016	Koba	B60L 50/64
2016/0294022	A1	10/2016	Thramann et al.	
2017/0301961	A1 *	10/2017	Kim	H01M 50/522
2017/0338519	A1	11/2017	Junger et al.	

2017/0361717	A1	12/2017	Qin et al.	
2018/0170201	A1	6/2018	Miller et al.	
2019/0135116	A1	5/2019	Narla	
2019/0168630	A1	6/2019	Mrlik et al.	
2019/0207269	A1 *	7/2019	Junger	H01M 10/4207
2020/0055405	A1	2/2020	Duan et al.	
2020/0067319	A1	2/2020	Qin et al.	
2020/0231064	A1 *	7/2020	Zhao	B60L 50/64
2020/0313249	A1 *	10/2020	Zhao	H01M 10/48
2021/0008995	A1	1/2021	Lee et al.	
2021/0265710	A1	8/2021	Muenzel et al.	
2021/0313630	A1 *	10/2021	Zeiler	H01M 10/482
2022/0032806	A1 *	2/2022	Christensen	B60L 53/62
2023/0101550	A1 *	3/2023	Bentzion	B60L 50/60 320/109
2023/0139770	A1 *	5/2023	Burchardt	H01M 10/425 429/50
2023/0402667	A1 *	12/2023	Warren	G01R 31/392

FOREIGN PATENT DOCUMENTS

JP	2013103723	A	5/2013	
JP	2017168244	A	9/2017	
WO	2006079202	A1	8/2006	
WO	2018158453	A1	9/2018	
WO	WO-2020051483	A1 *	3/2020	B60L 50/64
WO	2020077126	A1	4/2020	
WO	2021014363	A1	1/2021	
WO	2021019444	A1	2/2021	
WO	2022006159	A1	1/2022	

OTHER PUBLICATIONS

Bauer, Michaela et al., "Evaluating frequency regulation operated on two stationary energy systems with batteries from electric vehicles", Science Direct, Energy Procedia 155, 2018, pp. 32-43.

Ahmadi, Leila et al., "A cascaded life cycle: reuse of electric vehicle lithium-ion battery packs in energy storage systems", Int J Life Cycle Assess, Sep. 7, 2015, pp. 111-124.

Williams, Dexter M.T. J. et al., "Repurposing Used Electric Vehicle Batteries for Energy Storage of Renewable Energy in the Power System", University of Manitoba, Manitoba HVDC Research Centre, 2012 25th IEEE Canadian Conference on Electrical and Computer Engineering (CCECE), 6 pages.

Extended European Search Report, issued on Nov. 19, 2024, in corresponding European Application No. 22729018.6-1002, 10 pages.

Communication pursuant to Rules 70(2) and 70a(2) EPC, issued Dec. 6, 2024, in corresponding European Application No. 22729018.6-1002, 1 page.

Notice of Reasons for Refusal, issued Jul. 9, 2024, in corresponding Japanese Application No. 2023-560256, 11 pages.

Communication pursuant to Rules 161(2) and 162 EPC, issued Sep. 25, 2024, in corresponding European Application No. 23756911.6-1009, 3 pages.

International Preliminary Report on Patentability, issued Aug. 20, 2024, in corresponding International Application No. PCT/US2023/013297, 5 pages.

* cited by examiner

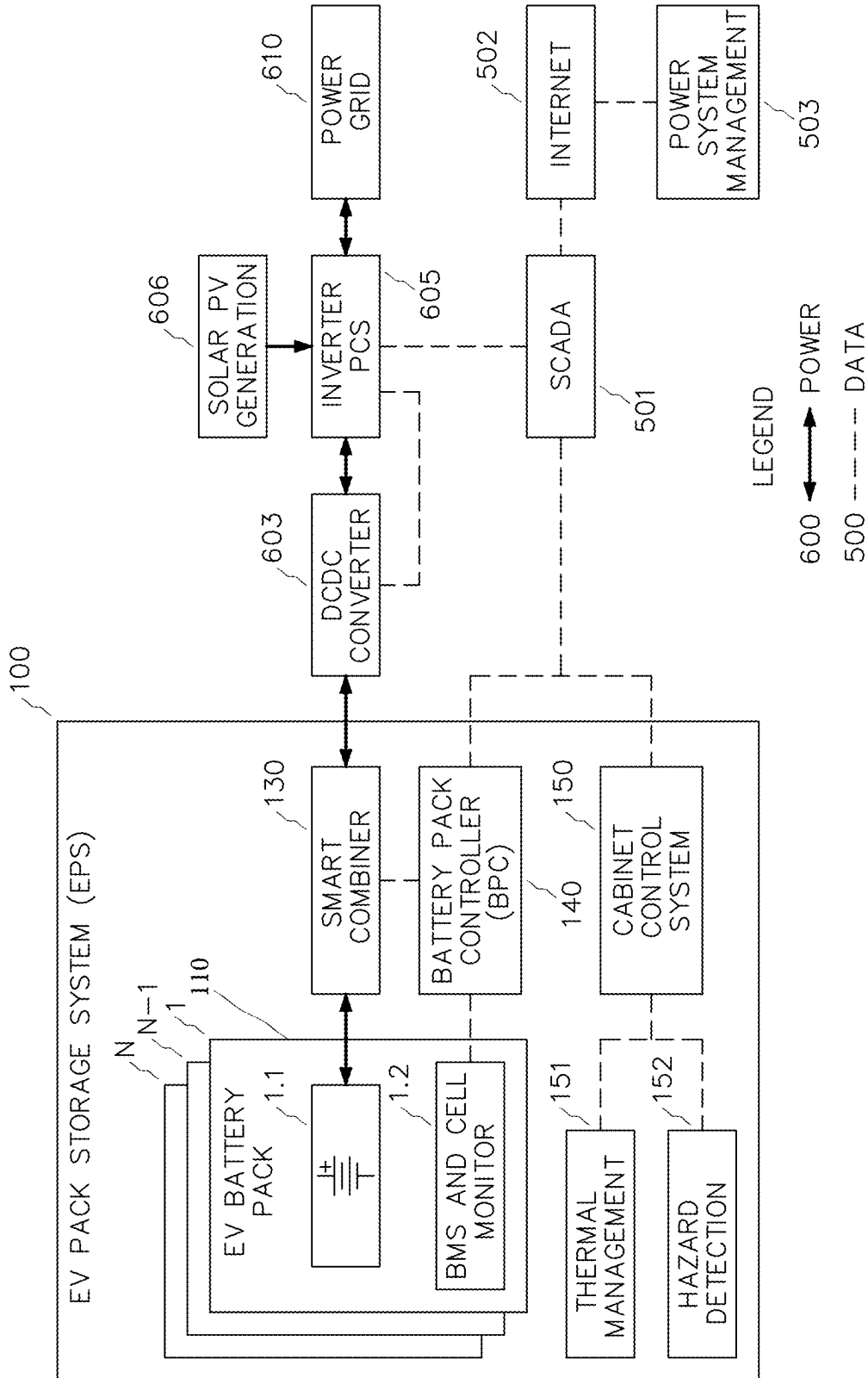


FIGURE 1

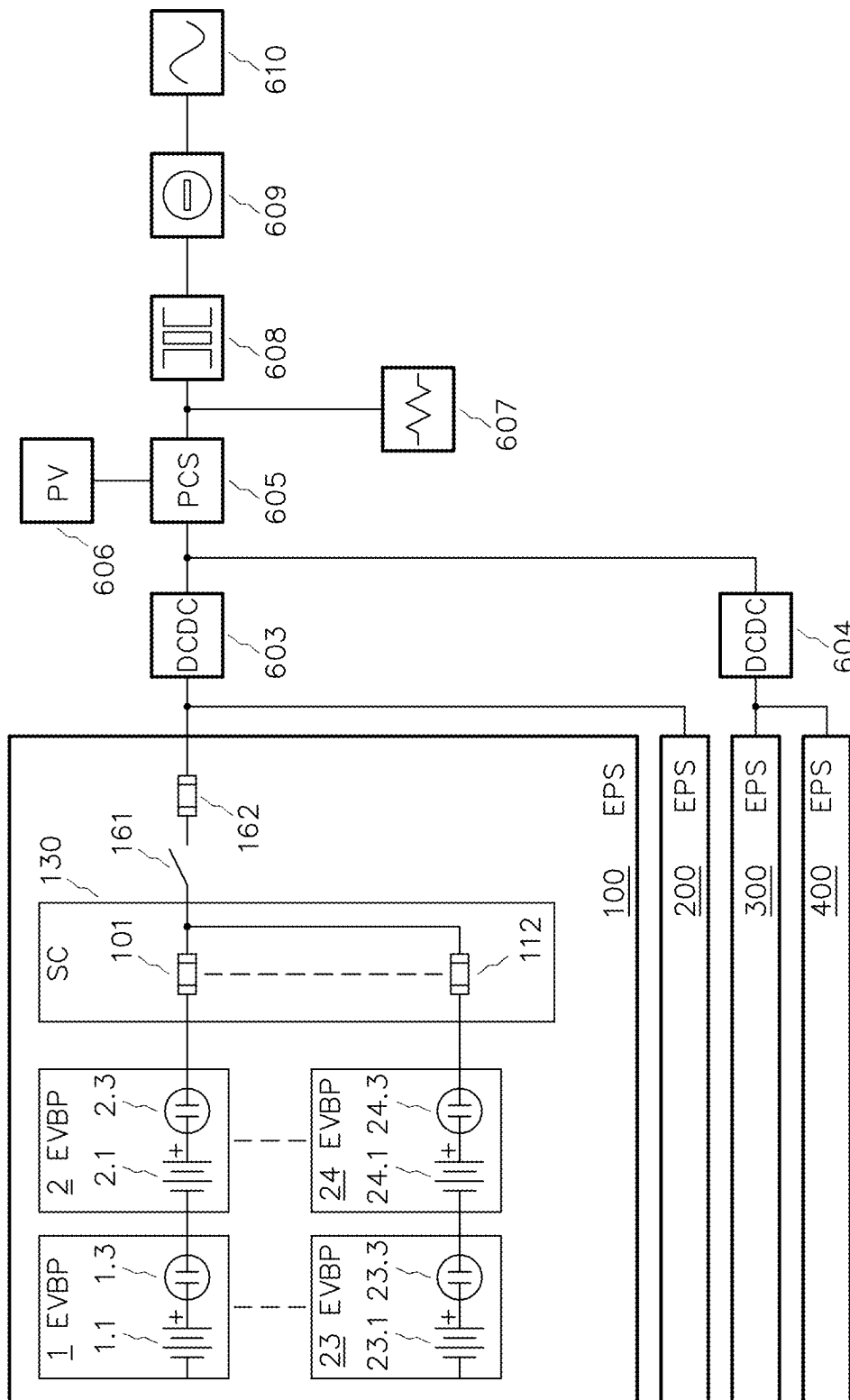


FIGURE 2

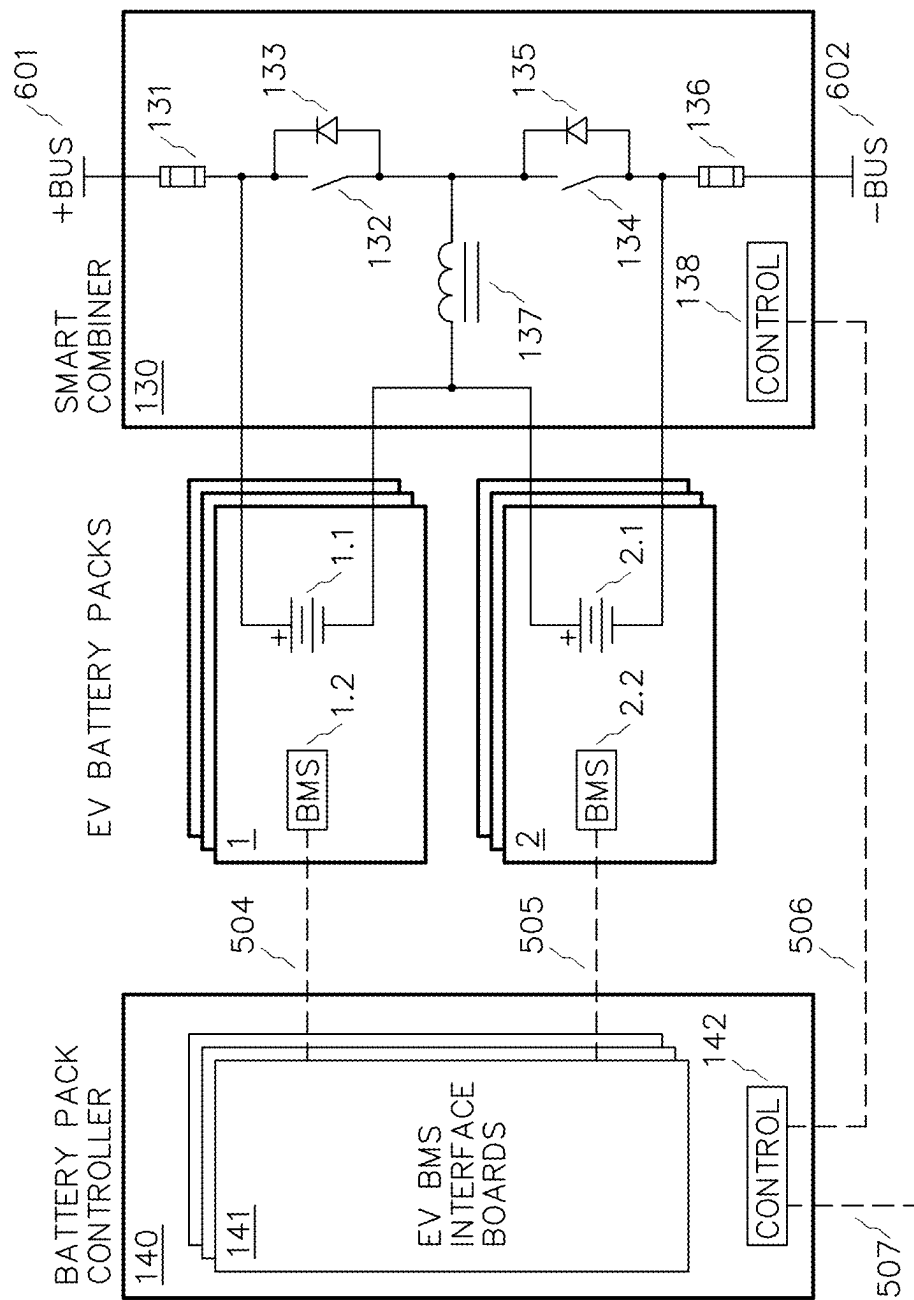


FIGURE 3

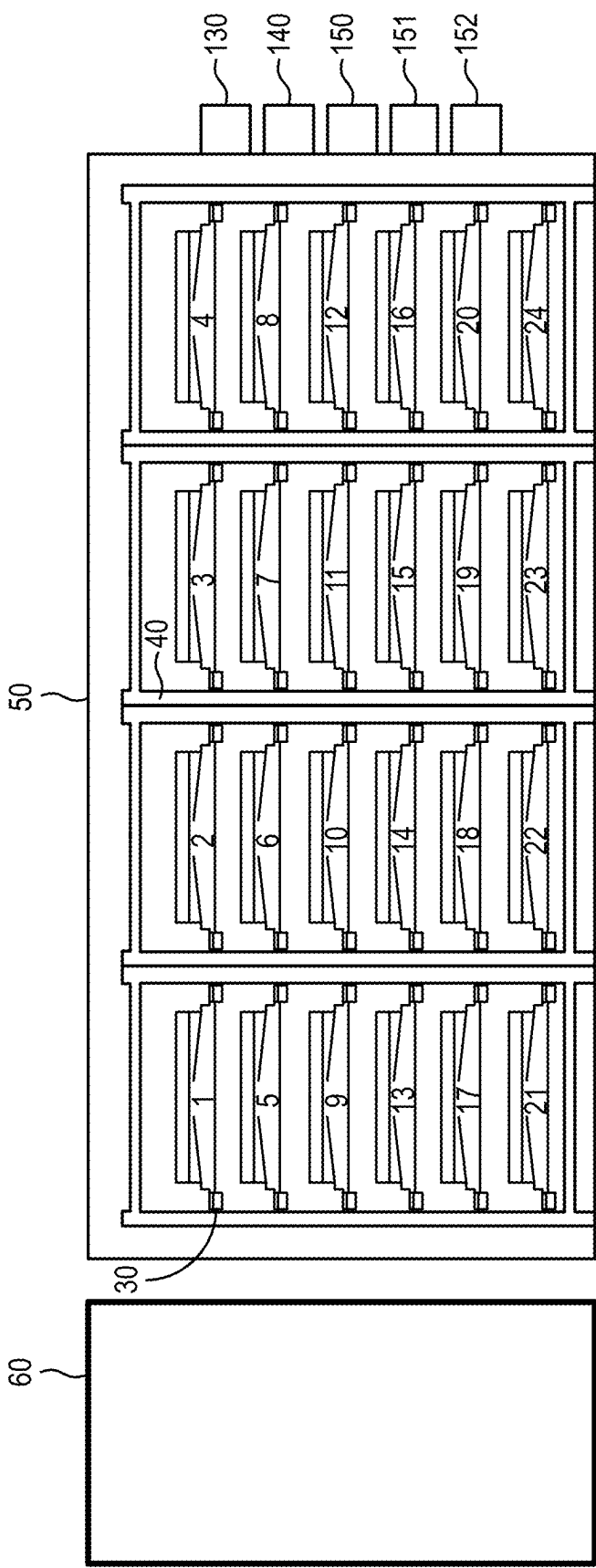
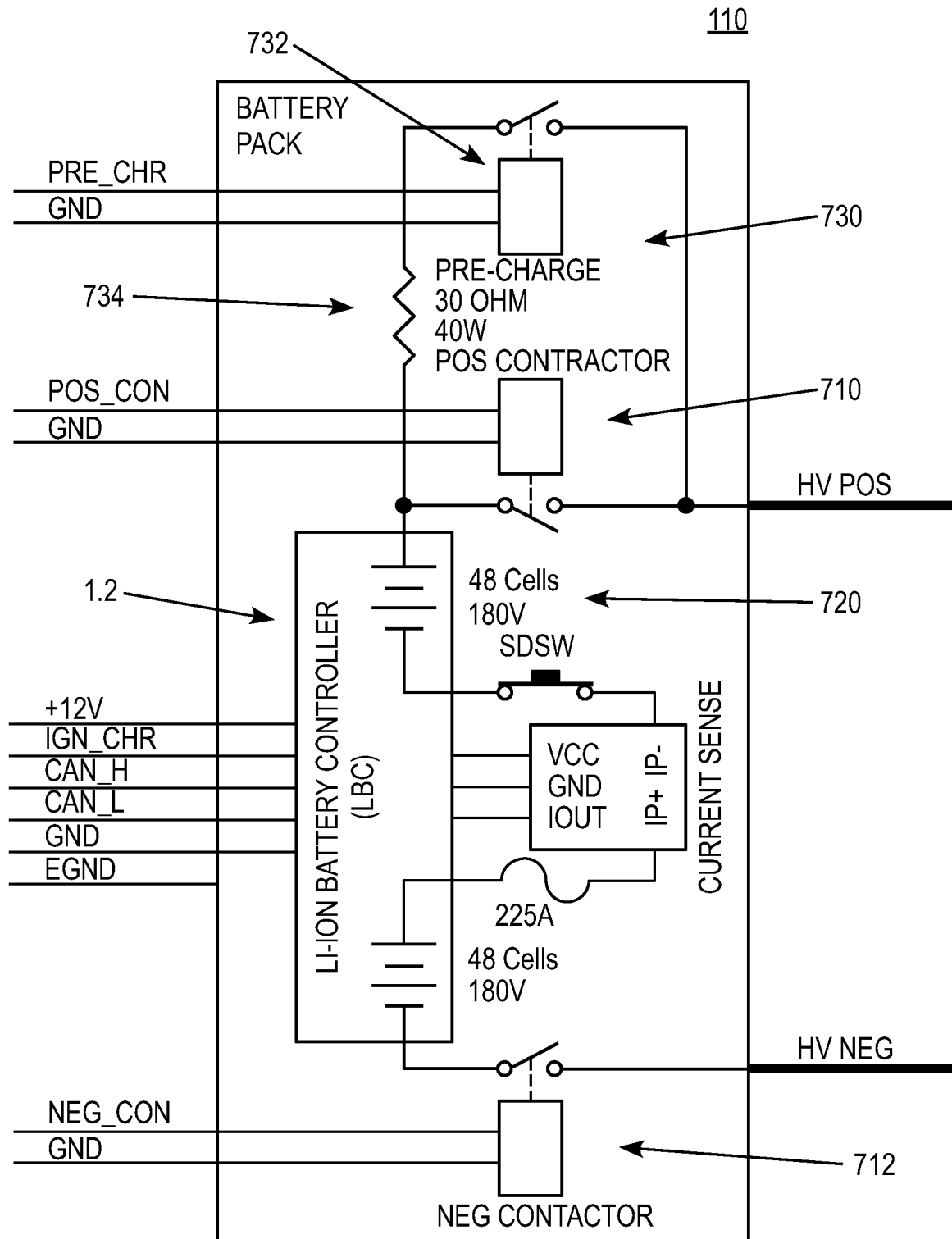


FIGURE 4

FIG. 5



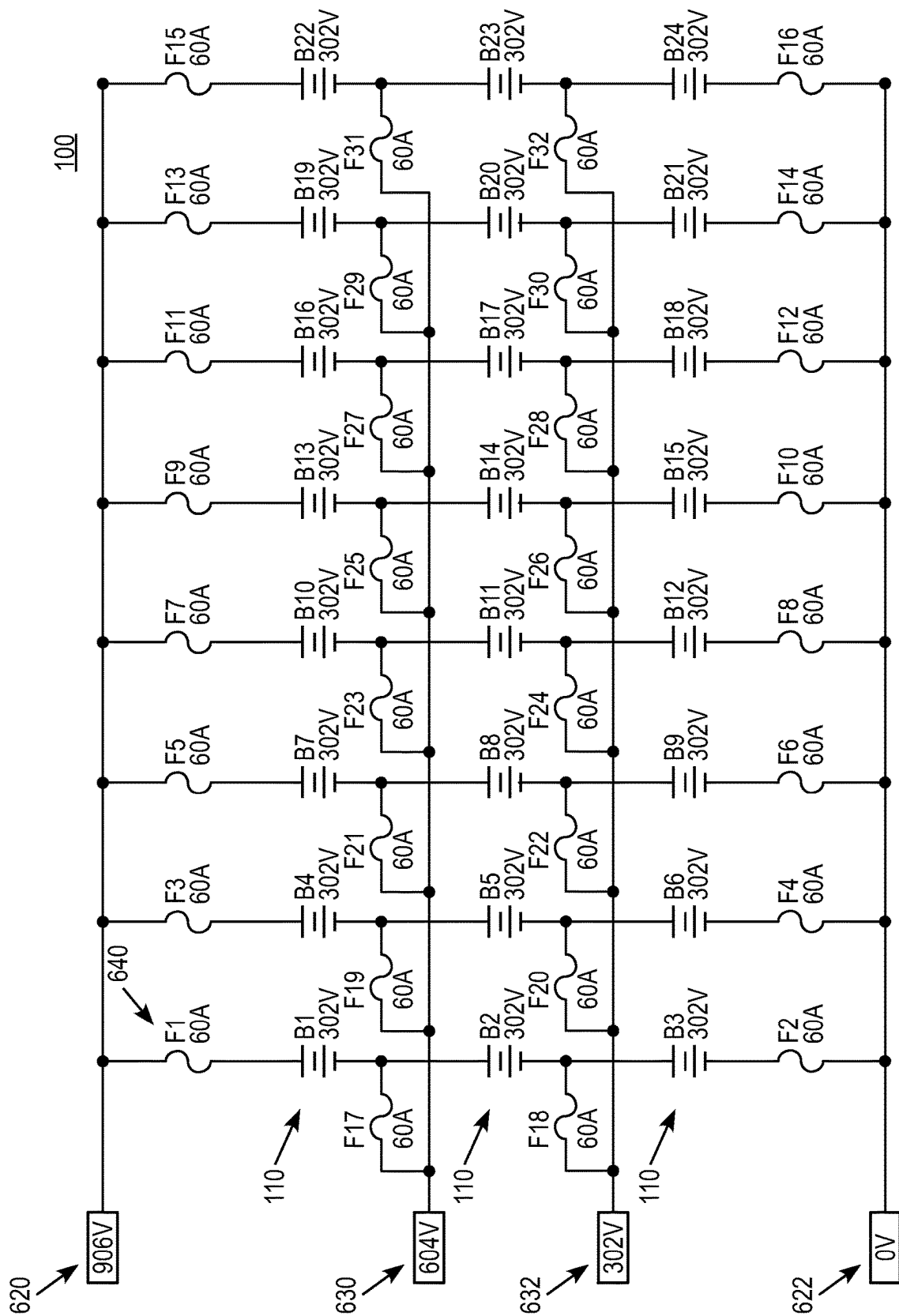


FIG. 6

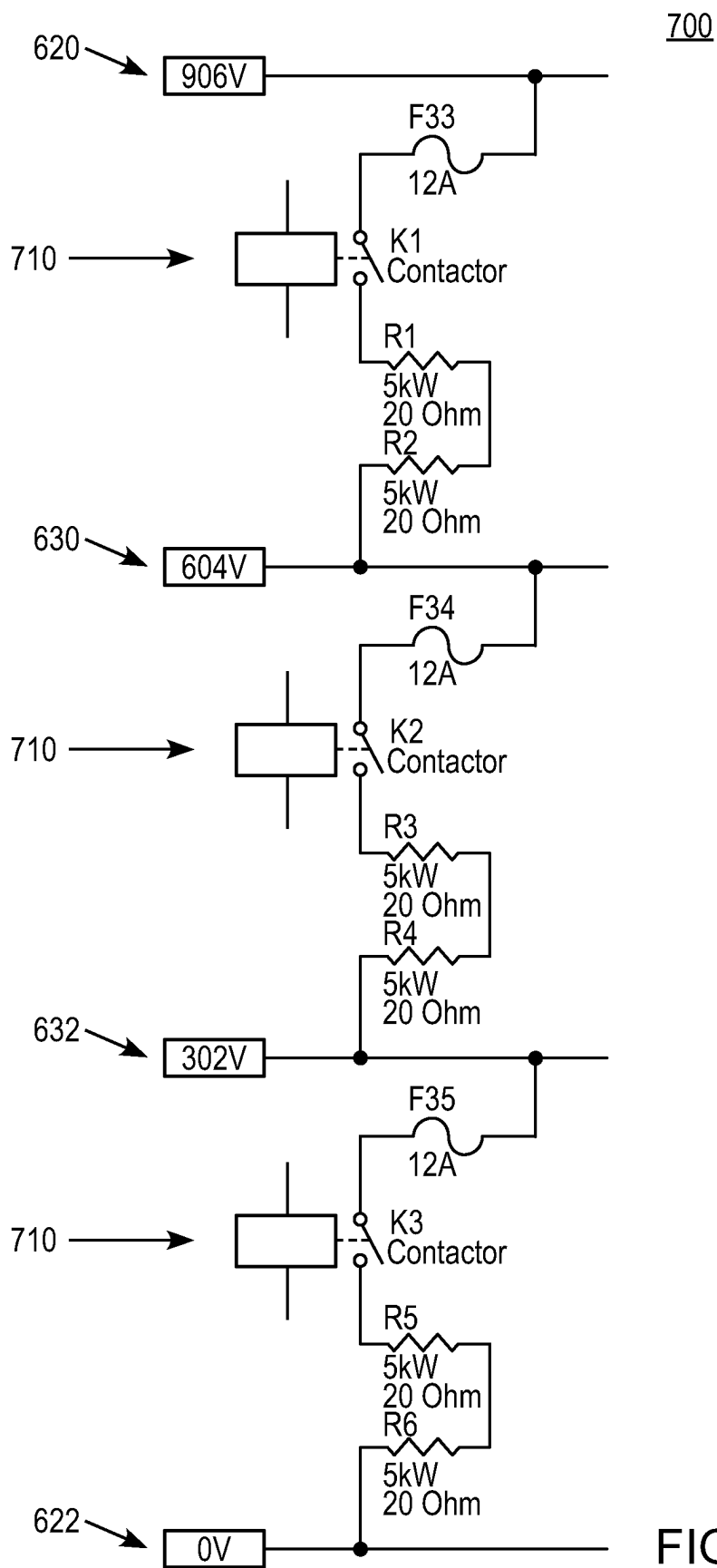
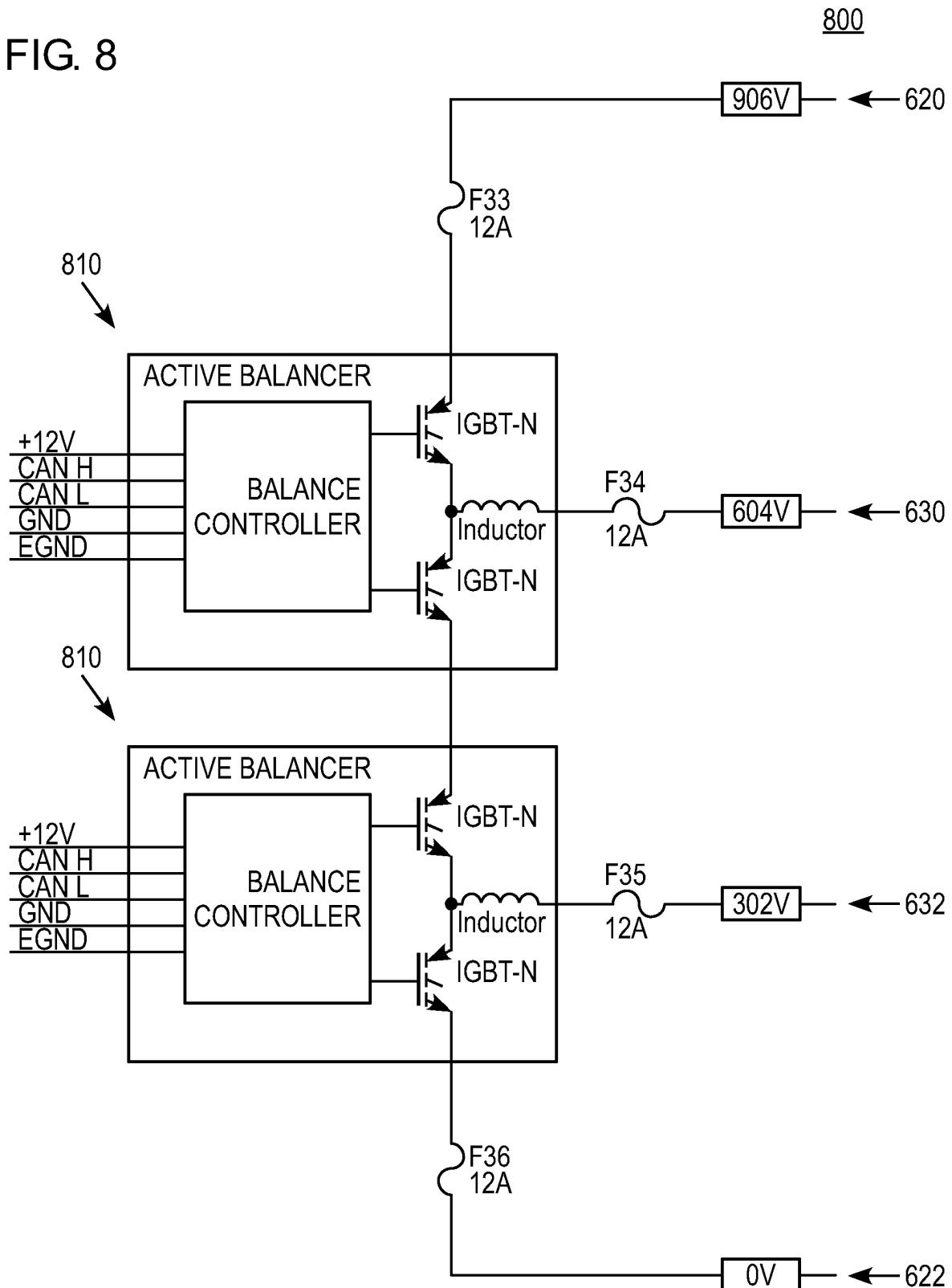


FIG. 7

FIG. 8



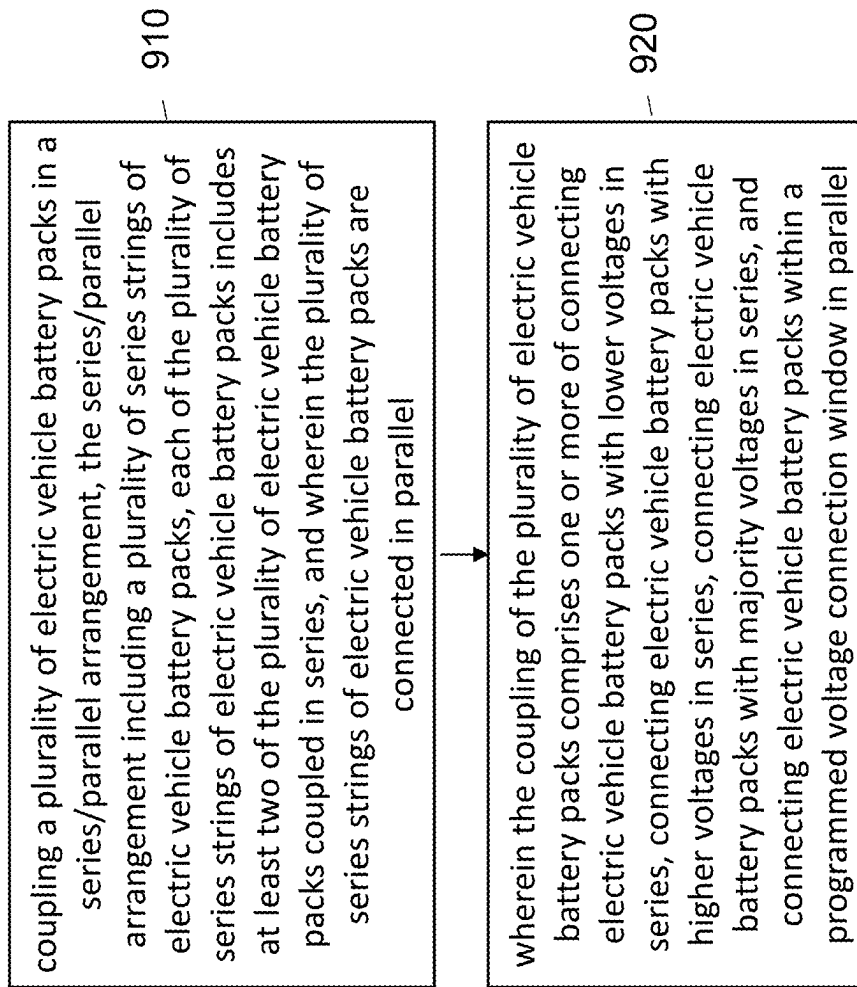
900

FIGURE 9

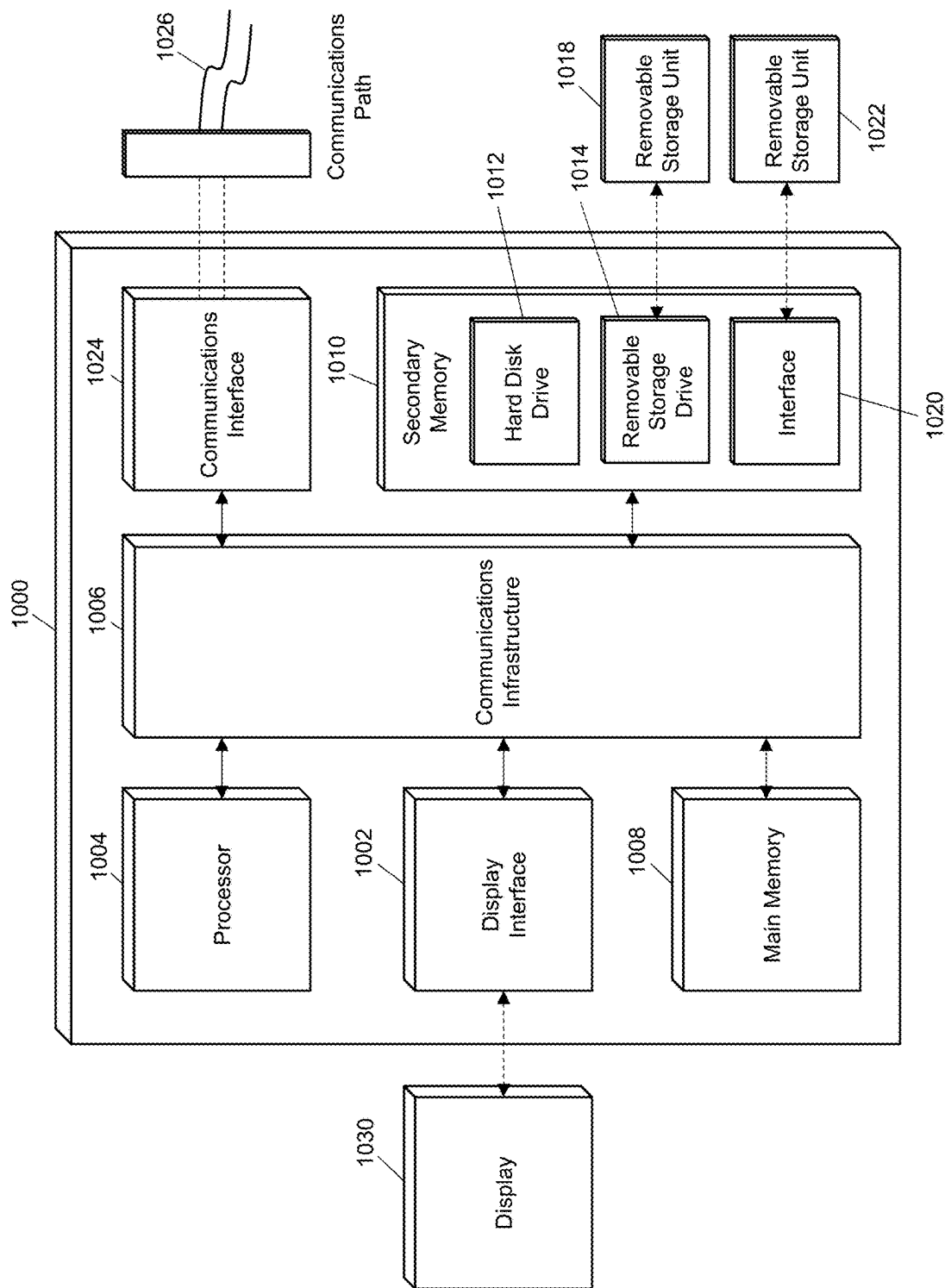


FIGURE 10

1

ENERGY STORAGE SYSTEM AND METHOD EMPLOYING SECOND-LIFE ELECTRIC VEHICLE BATTERIES

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation-in-part of U.S. application Ser. No. 17/382,933 filed on Jul. 22, 2021, which is a continuation of U.S. application Ser. No. 17/118,497 filed on Dec. 10, 2020, which are both incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present disclosure generally relates to an energy storage system and method employing second-life electric vehicle batteries, and more particularly to an integrated battery energy storage system, which includes a plurality of electric vehicle battery packs coupled in a series/parallel arrangement and maximizing yields from the plurality of electric vehicle battery packs with varying states of health (SOH).

BACKGROUND

Reducing the cost of energy storage systems (ESS) is an important objective for electricity ratepayers and policy-makers. A primary metric to measure ESS cost is the levelized cost of storage (LCOS), defined as the total lifetime cost of the ESS, including capital costs to construct as well as costs to operate, divided by the cumulative delivered electricity that the system has stored.

Repurposing electrochemical batteries from electric vehicles (EV) for a second use or second-life stationary storage application can significantly reduce the LCOS of ESS compared to using new batteries. Deploying EV batteries in stationary storage applications is the highest and best use of batteries when the batteries are no longer suitable for use in EVs. This in turn, allows a 2nd life battery pack to return many times more than its initial investment before being broken down for component recycling. Larger scale ESS requires a large number of batteries to be deployed in series and parallel electrical configurations to deliver energy at high voltage and current levels. Whether electrically connected in front of or behind a customer's meter, an ESS must efficiently integrate and manage batteries over time and charge/discharge cycles to be effective.

SUMMARY

It would be desirable to have a system that utilizes a plurality of electric vehicle (EV) batteries in second-life stationary storage applications within an overall energy storage system (ESS).

In accordance with an aspect, an integrated battery energy storage system is disclosed, the integrated battery energy system comprising: a plurality of electric vehicle battery packs; and a computer system, the computer system including a processor configured to: couple the plurality of electric vehicle battery packs in a series/parallel arrangement, the series/parallel arrangement including a plurality of series strings of electric vehicle battery packs, each of the plurality of series strings of electric vehicle battery packs includes at least two of the plurality of electric vehicle battery packs coupled in series, and wherein the plurality of series strings of electric vehicle battery packs are connected in parallel;

2

and wherein the coupling of the plurality of electric vehicle battery packs includes one or more of connecting electric vehicle battery packs with lower voltages in series, connecting electric vehicle battery packs with higher voltages in series, connecting electric vehicle battery packs with majority voltages in series, and connecting electric vehicle battery packs within a programmed voltage connection window in parallel.

In accordance with another aspect, a method is disclosed for integrating electric vehicle battery packs into an integrated battery energy storage system, the method comprising: coupling a plurality of electric vehicle battery packs in a series/parallel arrangement, the series/parallel arrangement including a plurality of series strings of electric vehicle battery packs, each of the plurality of series strings of electric vehicle battery packs includes at least two of the plurality of electric vehicle battery packs coupled in series, and wherein the plurality of series strings of electric vehicle battery packs are connected in parallel; and wherein the coupling of the plurality of electric vehicle battery packs comprises one or more of connecting electric vehicle battery packs with lower voltages in series, connecting electric vehicle battery packs with higher voltages in series, connecting electric vehicle battery packs with majority voltages in series, and connecting electric vehicle battery packs within a programmed voltage connection window in parallel.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating the utility as part of an energy storage system in accordance with an exemplary embodiment.

FIG. 2 is a modified electrical single-line schematic diagram of an energy storage system in accordance with an exemplary embodiment.

FIG. 3 illustrates a battery pack controller (BPC) and smart combiner (SC), with active balancer in accordance with an exemplary embodiment.

FIG. 4 illustrates the mechanical layout of an enclosure component in accordance with an exemplary embodiment.

FIG. 5 illustrates an example of internal components of an electric vehicle (EV) battery pack in accordance with an exemplary embodiment.

FIG. 6 illustrates an example of an energy storage system having three (3) electric vehicle battery packs coupled in a series string and eight (8) series strings of electric vehicle battery packs connected in parallel in accordance with an exemplary embodiment.

FIG. 7 illustrates an example of an external passive balancer for three (3) electric vehicle battery packs arranged in a series string in accordance with an exemplary embodiment.

FIG. 8 illustrates an example of an external active balancer for three (3) electric vehicle battery packs arranged in a series string in accordance with an exemplary embodiment.

FIG. 9 illustrates a method for integrating electric vehicle battery packs into an integrated battery energy storage system.

FIG. 10 illustrates an exemplary hardware architecture for an embodiment of a computer system.

DETAILED DESCRIPTION

For simplicity and illustrative purposes, the principles of the embodiments are described by referring mainly to

examples thereof. In the following description, numerous specific details are set forth in order to provide a thorough understanding of the embodiments. It will be apparent however, to one of ordinary skill in the art, that the embodiments may be practiced without limitation to these specific details. In some instances, well known methods and structures have not been described in detail so as not to unnecessarily obscure the embodiments.

In accordance with an aspect, an EV pack storage system (EPS) employs the EV battery packs as an integrated functional unit or building block wherein a plurality of EV battery packs are rather easily aggregated to behave as a larger battery within an ESS. The batteries can be racked in a specialized environmentally controlled enclosure in the original pack casing in which the batteries were mounted in the EV. The specialized environmentally controlled enclosure functions as an integrated EV pack storage unit and the EPS functions as a sub-system building block within the overall ESS. The EV battery packs within the EPS may be electrically connected in series as well as in parallel. Each series and parallel string are protected with an overcurrent device. The EPS enclosure is designed for easy installation, removal and replacement of EV battery packs. Each battery pack and string may be monitored with a proprietary battery pack controller (BPC). The BPC helps ensure proper operating parameters and monitors the health of each EV battery pack. The battery pack controller (BPC) can monitor the health of each EV battery pack by interfacing with the EV pack integrated battery management system (BMS). The BPC manages a smart combiner (SC) to actively or passively balance the second-life batteries in order to effectively utilize the charge and discharge cycle of individual packs adjusting for variance in each pack's capacity. The environment of the EPS enclosure is managed to maintain suitable operating temperatures, and for hazard detection. The EPS operates within a larger ESS that also includes a Power Conversion System (PCS) and Supervisory Control and Data Acquisition (SCADA) system. Multiple EPS can operate in integrated fashion together within an ESS. The ESS can be configured as DC-coupled, or AC-coupled to the inverter. The EPS can be charged from on-site generation such as solar or wind, or from electricity provided by the AC power system, for example, a power grid. The EPS within the ESS can be deployed in front of the meter (IFM) directly interconnected to the power grid or deployed behind the meter (BTM) to offset a customer's load and demand.

In accordance with an exemplary embodiment, an integrated system is disclosed for deploying a plurality of second-life electric vehicle (EV) battery packs within an energy storage system (ESS).

In the present disclosure, herein referred to as the EV pack storage system (EPS), is an integrated functional building block wherein a number of EV battery packs are easily aggregated to behave as a larger battery within an overall ESS. The EPS composition and function can include: (1) the EV battery packs are utilized, both mechanically and electrically, as they were in the original first-life vehicle application, incorporating the battery pack's battery management system (BMS) as well as a similar digital serial data link format and protocol; (2) the EV battery packs are electrically configured in a parallel arrangement and often in series. EV battery packs may not have been originally designed for use in series and/or parallel connection but this limitation is overcome with a unique design for mounting, communicating and interconnecting; (3) a number of EV battery packs are integrated into an environmentally controlled and monitored cabinet or enclosure which is not considered an

occupiable space according to building or fire code definitions, and where a single enclosure, or multiple enclosures, can be integrated in the ESS; (4) a battery pack controller (BPC) is deployed as part of the EPS to integrate the communications and controls necessary for the batteries to work together and operate in coordination as a unified functional block; and (5) a smart circuit combiner provides electrical balancing and overcurrent protection for all EV battery packs within the EPS.

FIG. 1 is a block diagram illustrating the utility of the disclosure in a preferred embodiment as part of a grid-tied energy storage system. In accordance with an exemplary embodiment, the EV pack storage system (EPS) **100** can be implemented as part of an overall energy storage system (ESS). All heavy lines **600** with arrowheads indicate power connections and possible power flow directions. All dashed lines **500** indicate bidirectional digital data bus connections.

EPS **100** includes a plurality of EV battery packs **110**, designated in FIG. 1 as blocks **1**, **N-1** and **N** to indicate any number **N** of identical EV battery packs. EV battery pack **1** contains batteries **1.1** and Battery Management System (BMS) **1.2**. The EV battery pack batteries **1.1** through **N.1** are connected in series at nominal voltages, which are multiples of a single pack's nominal voltage. A number **N** of EV battery packs **110** are connected in a series/parallel configuration or otherwise aggregated within smart combiner **130** to electrically behave as a larger battery within the overall energy storage system. A smart combiner **130** is connected to a bidirectional DCDC (DC-to-DC) converter **603**. A DCDC converter **603** can provide an optimum voltage match between the aggregate EV battery packs and the Inverter Power Conversion System (PCS) **605** to provide an optimum voltage match as the maximum power point of solar PV generation **606** changes with temperature and load. A smart combiner (SC) **130** can be used to actively balance the second-life batteries in order to effectively utilize the charge and discharge cycle of individual packs adjusting for variance in each pack's capacity.

When the overall system is delivering energy stored in the EV battery packs, for example, to a power grid **610**, DC power flows from the EV battery packs, through smart combiner **130** and through DCDC converter **603**. DC power is then converted to AC power by inverter PCS **605** to supply energy to the power grid **610**. Power from solar photovoltaic generation **606**, when available, for example, can also flow through PCS **605**, which functions as a DC to AC power converter to discharge power into power grid **610**. In this configuration, the total power into power grid **610** can be, for example, a combination of battery sourced power and solar photovoltaic sourced power.

In accordance with an exemplary embodiment, when the overall system is delivering energy to charge EV battery packs **1** through **N**, PCS **605** functions as an AC to DC power converter by sourcing AC power from power grid **610** and converting it to DC power. This DC power flows through DCDC converter **603**, through smart combiner **130** and into all EV battery packs **1** through **N**. Power from solar photovoltaic generation **606**, when available, can either be used to reduce the power required from power grid **610** to charge EV battery packs **1** through **N** or if the power from available solar power generation **606** is greater than the power required to charge these EV battery packs, then the excess power can be delivered to power grid **610**. Some system variants will not include solar PV onsite generation **606** and therefore DCDC converter **603** may not be required.

In accordance with an exemplary embodiment, a battery management system (BMS) **1.2** monitors every cell in the

5

EV battery pack **1**, primarily to check for mismatched, undercharged or overcharged cells in a series string. In addition the BMS can also monitor pack voltage, current and temperature. By monitoring and rebalancing mismatched cells, limiting current, voltage and temperature, the usable lifetime of the EV battery pack can be enhanced, and battery cell and pack safe operating parameters can be ensured to avoid hazardous conditions. A Battery Pack Controller (BPC) **140** communicates with individual EV battery packs **1** through **N** over a digital data bus. Supervisory Control and Data Acquisition (SCADA) **501** communicates with BPC **140** to ascertain the state of charge, state of health and overall availability of the aggregate EV battery packs. In accordance with an exemplary embodiment, SCADA **501** receives top level commands from power system manager **503** via internet **502** to control the operation of the overall energy storage system (EPS) **100, 200, 300, 400** (FIG. 2).

SCADA **501** can also communicate with a cabinet control system **150**. Within EPS **100**, the cabinet control system **150** can communicate with thermal management block **151** and the hazard protection block **152**. Thermal management block **151** provides air conditioning, dehumidification, venting, and air circulation as required to maintain an optimum environment for EV battery packs **1** through **N**. The hazard detection block **152** monitors the environment inside the EPS **100** enclosure, for example, for smoke and over/under temperature conditions.

FIG. 2 is a modified single-line electrical diagram of an energy storage system in accordance with an exemplary embodiment and illustrates the power flow. This disclosure and description does not limit, however, the scope of the disclosure to systems with the number of elements and/or components described herein.

As illustrated in FIG. 2, the energy storage system, for example, has four EV pack storage assemblies (EPS) **100, 200, 300** and **400**. Each EPS **100, 200, 300, 400**, can be identical. However, each EPS may not be identical and modifications of one or more the EPSs **100, 200, 300, 400** may occur. In accordance with an exemplary embodiment, the EPS **100** contains 24 EV battery packs (EVB) designated **1-24**. Within each EVBP are a number of series or series parallel connected battery modules shown as **1.1** in EVBP **1** through **24.1** in EVBP **24**. There are also a normally open contactors, which may be configured as one or more series connected contactors designated, **1.3** in EVBP **1** through **24.3** in EVBP **24**. EVBP **1-24** may also contain other power circuits, such as but not limited to pre-charge resistors, relays and fuses. In this exemplary system, EVBP **1** through **24** are connected in a 2S12P circuit arrangement where EVBP pairs are connected in series and where 12 of these series strings are connected in parallel. The series strings pairs are **1/2, 3/4, 5/6, 7/8, 9/10, 11/12, 13/14, 15/16, 17/18, 19/20, 21/22, and 23/24**. Each of the 12 series strings **1/2** through **23/24** are connected through fuses **101** through **112**, respectively, in smart combiner **130**. Fuses **101** through **112** provide a second tier of overcurrent protection to protect the EV batteries **1.1** through **24.1** from over-currents under abnormal conditions. Fuses **101** through **112** can also prevent contactors **1.3** through **24.3** from breaking currents outside safe contactor limits. Contactors **1.3** through **24.3** can provide the first tier of overcurrent protection, for currents less than fuse **101** through **112** ratings, as well as other circuit isolation and connection control functions. The output of smart combiner (SC) **130** is connected to switch **161**, which connects or isolates the power circuits of EPS **100** from the other energy storage system components. Fuse **162** protects the ampacity of conductors and the break

6

capability of switch **161**. In accordance with an exemplary embodiment, circuit breakers or other overcurrent devices and systems may be substituted for fuses in FIG. 2.

In this system example, EPS **100** and **200** are connected to DCDC converter **603**. EPS **300** and **400** are connected to DCDC converter **604**. DCDC converter **603** can provide an optimum voltage match between parallel connected EPS **100** and EPS **200** and the inverter power conversion system (PCS) **605** to provide an optimum voltage match as the maximum power point of solar PV generation **606** changes with temperature and irradiance. Both DCDC converters **603** and **604** are capable of bidirectional power transfer, to either charge or discharge EV battery packs. PCS **605** is also bidirectional with respect to power flow.

When the overall system, for example, is delivering energy stored in EPS **100, 200, 300** and **400** batteries to power grid **610** and/or local loads **607**, DC power flows from EPS **100** and **200** through DCDC converter **603** and in parallel from EPS **300** and **400** through DCDC power converter **604**. DC power can then be converted to AC power by PCS **605**. AC power then flows through distribution transformer **608**, where the voltage can be raised to more efficient distribution voltage levels and through revenue meter **609** before connection to power grid **610**. Power from solar photovoltaic generation **606**, when available and utilized, also flows through PCS **605** and PCS **605** functions as a DC to AC power converter to source power into power grid **610** and/or local loads **607**. The total power into power grid **610** can be the sum of battery sourced power plus solar photovoltaic sourced power minus the power used by local loads **607**. Local loads may also be supported without a connection to power grid **610** in which case PCS **605** works in a "stand alone" AC voltage regulation mode.

When the overall energy storage system is delivering energy to charge EV battery packs in EPS **100, 200, 300** and **400**, PCS **605** functions as an AC to DC power converter by sourcing AC power from power grid **610** and converting AC to DC power. In accordance with an exemplary embodiment, the DC power flows through DCDC converters **603** and **604** to charge EV battery packs in EPS **100, 200, 300** and **400**. Power from solar photovoltaic generation **606**, when available, for example, can either be used to reduce the power required from the power grid **610** to charge EV battery packs, or if the power available from solar power generation **606** is greater than the power required to charge these EV battery packs and supply local "behind the meter" loads **607**, then the excess power can be delivered to power grid **610**.

In accordance with an alternate embodiment as shown in FIG. 2, the energy storage system can include one or more of the following: PCS **605**, DCDC converter **603**, transformer **608** and power meter **609**, as part of the integrated EPS **100**.

FIG. 3 is a detailed drawing of the battery pack controller (BPC) **140** and smart combiner **130** as shown in FIG. 1. Reference designator **1** indicates a plurality of EV battery packs (EVB) with EVBP **1** on top. Reference designator **2** indicates a plurality EVBP with EVBP **2** on top. EVBP **1** and **2** are electrically connected in series as the first series pair in this EPS embodiment. Each series pair within the EPS communicates with one EV BMS interface board (EVIB) within in the BPC **140**. In this first, top level case, BMS **1.2** and BMS **2.2** communicate with EVIB **141** over digital data buses **504** and **505**, respectively. The communication protocol is the same as used in the original manufacturer's electric vehicle application. In most instances, but not limited to controller area network bus (CAN bus). All other EVBP series pairs within the EPS communicate in a similar

way. Control circuit **142** aggregates the data from all EVIB within BPC **140** and provides a Modbus digital communication link **507** to a system controller external to the EPS and Modbus link **506** to smart combiner **130**. In accordance with an exemplary embodiment, the Modbus digital communication link **507** communicates with the overall energy system SCADA (element **501** in FIG. 1) to report the voltage, current, health, availability and state of charge of the “composite” battery configured from the series/parallel connection of all EV battery packs. Modbus data link **506** connects control circuit **142** to control circuit **138** to actively control the balancing of each EV battery pack in a series string.

In accordance with an exemplary embodiment, the balancing of each EV battery pack in a series string, for example, can be achieved by monitoring the voltage, current and temperature of each EV battery pack and then transferring energy from the higher voltage of the two packs in a series string to the lower voltage pack. Using the example shown in FIG. 3, if EVBP **1** has a higher voltage than EVBP **2**, then semiconductor switch **132** is closed, current flows from battery **1.1** through inductor **137**, semiconductor switch **132** is opened and the current through inductor **137** flows through diode **135** and into battery **2.1** completing one energy transfer cycle. This energy transfer cycle is repeated at high frequencies for an amount of time proportional to the desired energy transfer. If EVBP **2** has a higher voltage than EVBP **1**, then semiconductor switch **134** is closed, current flows from battery **2.1** through inductor **137**, semiconductor switch **134** is opened and the current through inductor **137** flows through diode **133** and into battery **1.1** completing one energy transfer cycle.

FIG. 3 illustrates a method of balancing batteries in a series battery string but does not limit the disclosure to any one battery balancing method or number of EVBPs in a series string. In accordance with an exemplary embodiment, the energy storage system includes the interconnections and inter-functionality of the battery pack controller (BPC) **140**, EV battery pack battery management systems, **1.2** and **2.2** in this example, smart combiner **130** and overall system controller via data link **507**. Fuses **131** and **136** can be used to provide fault isolation between EVBP **1** and **2** series string and the parallel circuits of all series strings +BUS **601** and -BUS **602**. In accordance with an exemplary embodiment, all other series strings, for example, can be protected in a similar manner.

In accordance with an exemplary embodiment, the energy storage system can include each of the elements shown in FIG. 1 within EV pack storage system (EPS) **100** plus a weatherproof enclosure and a racking system for the EV battery packs. FIG. 4 illustrates an embodiment of the enclosure, which can include fixed and removable exterior insulated panels **60**. The enclosure **50** can be designed, for example, as a cabinet where the battery system and system controls are accessible from outside the enclosure, and the enclosure or cabinet is not an occupiable space as defined in building or fire codes. In accordance with an exemplary embodiment, the enclosure **50** is an outdoor rated enclosure that enables outdoor deployment of the energy storage assembly.

In accordance with an exemplary embodiment, the racking system **40** can be fabricated from structural steel and configured to hold EV battery packs **110** (packs **1** through **24**). Each pack **110** in the present build of the EV pack storage system (EPS), for example, can weigh in excess of 600 pounds. The side panels **60** of the enclosure **50** can be removed, for example, to allow forklift access to easily install and remove EV battery packs **1** through **24**. In

addition, the racking system **40** is designed to allow airflow between the EV battery packs. For example, each EV battery pack **110** can be electrically isolated from the racking system **40** by standoff **30**. The EV battery packs **110**, for example, can be cooled and heated by a thermal management system **151** under the direction of cabinet control system **150**. The thermal management system **151** may include cooling, dehumidification, and other environmental controls as a means for extending the useful lifetime of the EV battery packs **110**. A hazard detection system **152** can provide alarms to cabinet control system **150** when smoke, gas or temperature conditions outside of an operational range are detected.

The smart combiner **130** and battery pack controller (BPC) **140** functions are described in connection with the FIG. 1 narrative. The smart combiner **130**, the BPC **140**, the cabinet control system **150**, the thermal management system **151** and the hazard detection system **152** are not shown to scale or with specific physical features. In one or more of the smart combiner **130**, the BPC **140**, the cabinet control system **150**, the thermal management system **151** and the hazard detection system **152** can be mounted to the EPS enclosure. In accordance with an exemplary embodiment, the fixed and removable exterior panels **60** of the enclosure **50** may be thermally insulated.

In accordance with an exemplary embodiment, a system and method for improving yields from a plurality of electric vehicle (EV) battery packs **110** with varying state of health (SOH) are disclosed, which can help improve the storing and extracting of energy capacity from the plurality of electric vehicle (EV) battery pack systems **100**, **200**, **300**, **400**. As set forth, each of the EV battery packs **110** can be used as is, without modifications. In addition, battery cell module assemblies (CMA) comprising a plurality of cells that are assembled into a battery pack can also be used.

FIG. 5 illustrates internal components of an EV battery pack **110**. In accordance with an exemplary embodiment, each battery pack **110** contains a dedicated Battery Management System (BMS) **1.2**, for example, a Li-ion Battery Controller (LBC) that is provided by the manufacturer of the electric vehicle battery pack **110** or installed by an assembler if the electric vehicle battery pack is a battery cell module assembly (CMA). The BMS **1.2** can be configured to monitor the individual cell voltages and temperatures as well as total pack current and voltage and provides a communications method to report this information (for example, to a controller area network (CAN) bus). The BMS **1.2** can also maintain and balance the individual cells **720**, either actively or passively. In addition, as set forth above, the BMS **1.2** may also manage cooling/heating of the battery pack **110**. Each battery pack **110** can also contain one or more contractors **710**, **712**, for example, a positive contractor **710** and a negative contractor **712**, capable of connecting/disconnecting under load. Each battery pack **110** may also contain one or more pre-chargers **730** (power resistor **732** and relay **734**), which can be used to limit current when, for example, contractors are connecting/disconnecting the battery pack **110** under load. In addition, each of the battery packs **110** will also contain all the necessary electrical cables, cell interconnects, fuses, safety disconnects, connectors, components and housings to make them relatively safe to touch and rather easy to assemble into useable batteries (or battery storage systems).

As described above, the battery cabinets **50** can provide a secure weather protected and climate-controlled environment for the battery packs **110**. The battery packs **110** can be loaded and secured into cabinets **50** and wired into a

series/parallel matrix to form a battery. The battery packs **110** can be wired in series up to maximum working voltage of the inverter/DC-DC converter, which can result in two or more battery packs in series (2s, 3s, 4s . . . Ns). In accordance with an exemplary embodiment, any number of battery packs can be wired in parallel (2p, 3p, 4p . . . Np) to get the desired battery storage capacity. For example, FIG. **6** illustrates a battery matrix (e.g., a 3s8p matrix) comprising eight (8) series strings wired in parallel, and wherein each of the series strings of batteries comprises three (3) battery packs **110**.

As shown in FIG. **6**, each battery pack **110** can be individually wired and fused to form a positive rail **620** and negative rail **622**, along with intermediate rails **630**, **632** at the locations of each parallel battery connection (typically called a "center tap" in 2s configurations), there will be Ns-1 intermediate rails. For example, for a series string having three battery packs, the number of intermediate rails **630**, **632** will be two (2). The intermediate rails **630**, **632** can be used for series voltage balancing. In FIG. **6**, fuses **1-32** **610**, for example, can be in the combiner box (or smart combiner) **130** (FIG. **3**), which is attached, for example, to the side of the cabinet **50**. For example, the fuses **1-32** **640** can be sized to protect the wiring and the fuses should be rated more than the max charge/discharge current that the battery packs **110** will be exposed to in the matrix configuration of the electric vehicle battery pack storage system **100**. The combiner box **130** can also contain the main disconnect and main fuses for the cabinet **50**.

In accordance with an embodiment, internal to each battery pack **110** can be a cell balancer, which cell balancer can be either passive (resistive) or active (uses inductors or capacitors to store and move charge). These internal balancers will maintain the cell voltages to be within, for example, a few millivolts (mV) of each other.

In addition to the internal cell balancers, as shown in FIGS. **7** and **8**, the system can include an external string balancer system **700**, **800**. The external string balancer system **700**, **800** can be integrated into and/or attached to the combiner **130** and can be, for example, a passive balancer (resistive) **710**, or an active balancer **810** that uses inductors or capacitors to store and move charge). In accordance with an exemplary embodiment, the external string balancer **700**, **800** can maintain, for example, the intermediate rails **630**, **632** at the same voltage.

In accordance with an exemplary embodiment, the battery pack controller (BPC) **140** can include two active components, a pack interface (PI) card **141** and pack controller (PC) card (or control circuit) **142** as shown in FIG. **3**. In addition, the BPC **140** can include a passive back plane for communications between the pack interface card (PI) **141** and the pack controller (PC) card (or control circuit) **142**. The BPC **140** can be configured to communicate directly with the battery packs **110** and can include the inputs/outputs (I/O) to enable and charge/discharge the battery packs **110**. Each BPC **140** may communicate and control one or more battery packs **110**. The BPC **140** can have different communications methods and I/O as needed to interface with different manufacture of battery packs **110**. The BPC **140** can also simulate other components in the vehicle that may be required for the battery pack **110** to be enabled. For example, the BPC **140** can gather all the data available from the battery pack **110** and store the data into a battery pack data structure. For example, the battery pack data structure can be unique to each manufacture of battery pack. In accordance with an embodiment, the battery pack data structure can capture constant data sampled once on startup,

which can include, for example, battery pack make, model, serial number, etc. In addition, real time data can be sampled at a set interval, for example, during a set time-period, for example, every second, two seconds, etc. The real time data can include, for example, cell voltages, temperatures, pack voltage, current, state of charge (SOC), state of health (SOH), status, trouble codes, etc.

In accordance with an exemplary embodiment, the pack interface (PI) **141** can monitor the pack data and can set warnings or faults if the values are out of the expected ranges. For example, if there is a critical fault, or loss of communications, the battery pack **110** can be disabled by the BPC **140**.

In accordance with an exemplary embodiment, the BPC **140** communicates with each pack interface (e.g., battery management system (BMS) **1.2**) over a passive backplane and obtains a generic subset of the pack real time data. The pack real time data can be a condensed subset of data that can contain minimum and/or maximum cell voltages, temperatures and locations, pack voltage, current, state of charge (SOC), state of health (SOH), status warning and faults, etc. In accordance with an example, the BPC **140** can also monitor the pack data and sets any warnings or faults if the values are out of the expected ranges. If there is a critical fault, or loss of communications, the battery pack **110** can be disabled by the BPC **140**.

In addition, the BPC **140** can communicate to the supervisory control and data acquisition (SCADA) systems **501** over a native protocol of the BPC **140**, for example, over a TCP/IP Modbus. The BPC **140** can also communicate and coordinate with other battery pack controllers (BPCs) **140**, for example, a pack interface of the battery management system (BMS) **1.2**, connected to the same inverter/DC-DC converter. The BPC **140** commands each pack interface to enable/disable the battery packs **110** to connect/disconnect them from a battery matrix **100** as needed, for example, as set by a matrix algorithm.

In accordance with an embodiment, the EV battery packs **110** can be connected and disconnected under load. For example, the pre-charge resistor **730** (FIG. **5**) can be used to limit current before a pack **110** is connected or disconnected. When the pack Interface **141** receives a command to connect a pack **110**, the negative contactor is closed **712**, next the pre-charge resistor relay **730** is closed, then the positive contactor **710** is closed, finally the pre-charge resistor relay **730** is opened. When the pack interface **141** receives a command to disconnect a pack **110**. The pre-charge resistor relay **730** is closed, next the positive contactor **710** is opened, then the pre-charge resistor relay **730** is opened, finally the negative contactor **712** is opened.

In accordance with an exemplary embodiment, the BPC **140** can be commanded to connect the lowest voltage packs **110** (for example, to begin charging), connect highest voltage packs **110** (for example, to begin discharging), or connect a plurality of packs **110**, (for example, to connect as many packs **110** as possible for charge or discharge). When commanded to connect the lowest voltage packs **110**, a matrix algorithm is configured to locate the lowest voltage packs **110** to form a series string of battery packs **110**. The series string of battery packs, for example, two battery packs in series (2s), three battery packs in series (3s), four battery packs in series (4s) . . . Ns number of packs **110** can be told to connect, if successful this will become the first connected string. If one or more of the battery packs **110** faulted, another attempt will be made with the next lowest voltage non-faulted packs. Next, the SCADA system **501** is enabled

11

to charge the connected string, and wherein the charge current will be limited to the maximum programmed pack current.

In accordance with an exemplary embodiment, as the first series string starts to charge, the voltage of the connected packs 110 will begin to rise. When the voltage of one or more of the packs 110 of the first series string is within a programmed voltage connection window of the disconnected packs 110, the BPC 140 can enable one or more of the packs 110 to connect in parallel with the packs in the first series string having a detected voltage within the programmed voltage connection window of the disconnected packs. The pack interface 141 will then connect the packs 110 as instructed. As the packs 110 charge and the voltage rises, more disconnected packs 110 will match the programmed voltage connection window and connect. As more packs 110 are connected, the BPC 140 will adjust the SCADA 501 charge current to stay below the maximum programmed pack current. In accordance with an exemplary embodiment, eventually all non-faulted packs 110, for example, in the matrix 600 will be connected.

In accordance with an exemplary embodiment, the BPC 140 can be commanded to connect the highest voltage packs 110, the matrix algorithm will find the highest voltage packs to make a series string. For example, the series strings can be two voltage packs in series (2s), three voltage packs in series (3s), four voltage packs in series (4s) . . . Ns number of packs 110. The series strings of battery packs 110 can be instructed or programmed to connect, if successful this will become the first connected string of higher voltage packs. If one or more of the higher voltage packs faulted, another attempt will be made with the next highest voltage non-faulted packs 110 from the higher voltage packs 110. Next the SCADA system 501 can be enabled to discharge the connected series string, and wherein the discharge current will be limited to the maximum programmed pack current.

In accordance with an exemplary embodiment, as the first string of higher voltage packs 110 starts to discharge, the voltage of the connected packs 110 will begin to drop. When the voltage of one or more of the battery packs 110 is within the programmed voltage connection window of the disconnected packs 110, the BPC 140 will enable the one or more packs 110 with a voltage within the programmed voltage connection window to connect. The BPC 140 will then connect the battery packs 110 as instructed. As the battery packs 110 discharge and the voltage drops, more disconnected battery packs 110 will match the programmed voltage connection window and connect. As more battery packs are connected, the BPC 140 will adjust the SCADA 501 discharge current to stay below the maximum programmed pack current. In accordance with an exemplary embodiment, eventually all non-faulted packs in the matrix 600 can be connected.

In accordance with an exemplary embodiment, when commanded to connect the plurality of battery packs 110 that form a majority of the battery packs 110 that are within a programmed voltage connection window, the matrix algorithm can calculate a histogram of the plurality of battery packs 110 with binning determined by the programmed voltage connection window (e.g., the plurality of battery packs 110 with majority voltages that are within the programmed voltage connection window). The matrix algorithm can then make one or more series strings of the battery packs with the majority voltages, each of the one or more series strings having two or more battery packs in series (2s), three or more battery packs in series (3s), four or more battery packs in series (4s) . . . Ns number of packs, which

12

number of battery packs can be connect, and if successful, the battery packs 100 will become the first connected string. If one or more packs faulted, another attempt will be made with the other binned non-faulted packs 110. The remaining packs 110 within the programmed voltage connection window will be instructed to connect.

Next, the SCADA system 501 can be enabled to charge or discharge the connected series string, the charge/discharge current will be limited to the maximum programmed pack current. When any pack voltage gets within the programmed voltage connection window of the disconnected packs, the pack controller will enable those packs to connect. The pack interface will then connect the packs 110 as instructed. As more packs are connected, the pack controller will adjust the SCADA 501 charge current to stay below the maximum programmed pack current. In addition, once the battery packs with majority voltages (i.e., a majority of the packs are within a defined voltage range) have been connected in series, the series strings of electric vehicle battery packs in which the electric vehicle battery packs are within a programmed voltage connection window can be connected in parallel.

In accordance with an exemplary embodiment, due to the variance in battery pack state of health (SOH), capacity and individual cell balance, some battery packs 110 will become fully charged or fully discharge before others. For example, battery packs 110 reaching the maximum programmed cell voltage, and/or maximum state of charge (SOC) will be disconnected allowing the other battery packs 110 in the system to fully charge. Before disconnecting battery packs 110, the SCADA 501 charge current will be reduced as needed to stay below the maximum programmed pack current. While charging, battery packs 110 will eventually disconnect down to a programmed minimum number of connected series strings. The minimum number of connected strings is determined by the minimum inverter/DC-DC converter load. When the discharge begins, battery packs within the programmed voltage connection window will be instructed to connect and the discharge current will be limited to the maximum programmed pack current.

In accordance with an exemplary embodiment, EV battery packs 100, 200, 300, 400 reaching the minimum programmed cell voltage, and/or minimum state of charge (SOC) will be disconnected allowing the other battery packs 110 in the system to fully discharge. Before disconnecting packs 110, the SCADA 501 discharge current will be reduced as needed to stay below the maximum programmed pack current. While discharging, battery packs 110 will eventually disconnect down to a programmed minimum number of connected strings. For example, the minimum number of connected strings is determined by the minimum inverter/DC-DC converter load. When the charge begins, battery packs 110 within the programmed voltage connection window will be instructed to connect and the discharge charge will be limited to the maximum programmed pack current.

FIG. 9 illustrates a method 900 for integrating electric vehicle battery packs into an integrated battery energy storage system. As shown in FIG. 9, the method includes coupling a plurality of electric vehicle battery packs in a series/parallel arrangement, the series/parallel arrangement including a plurality of series strings of electric vehicle battery packs, each of the plurality of series strings of electric vehicle battery packs includes at least two of the plurality of electric vehicle battery packs coupled in series, and wherein the plurality of series strings of electric vehicle battery packs are connected in parallel 910. In addition, the

13

coupling of the plurality of electric vehicle battery packs comprises one or more of connecting electric vehicle battery packs with lower voltages in series, connecting electric vehicle battery packs with higher voltages in series, connecting electric vehicle battery packs with majority voltages in series, and connecting any other electric vehicle battery packs within a programmed voltage connection window in parallel **920**.

In accordance with an exemplary embodiment, the method can further include connecting the electric vehicle battery packs with lower voltages in series as at least a lower voltage series string; and charging the lower voltage series string to a maximum programmed pack current. In addition, wherein one or more of the electric vehicle battery packs with the lower voltages is faulted, further comprising: connecting one or more electric vehicle battery packs with a next lowest voltage to the lower voltage series string in place of the one or more the electric vehicle battery. The method can also include charging one or more of the electric vehicle battery packs connected in the lower voltage series string to a maximum programmed pack current of one or more disconnected electric vehicle battery packs; connecting in parallel one or more of the electric vehicle battery packs within a programmed voltage connection window of disconnected electric vehicle battery packs; and connecting one or more of the disconnected electric vehicle battery packs that are within the programmed voltage connection window of the disconnected electric vehicle battery packs to each other in one or more series strings.

In accordance with another exemplary embodiment, the method includes connecting the electric vehicle battery packs with higher voltages in series as a higher voltage series string; and discharging the higher voltage series string to a maximum programmed pack current. In addition, wherein one or more of the electric vehicle battery packs with the higher voltages is faulted, the method further includes: connecting one or more electric vehicle battery packs with a next highest voltage to the higher voltage series string in place of the one or more the electric vehicle battery packs with the higher voltages that is faulted. The method can also include discharging one or more of the electric vehicle battery packs connected in the higher voltage series string to a maximum programmed pack current of one or more disconnected electric vehicle battery packs; connecting in parallel one or more of the electric vehicle battery packs within a programmed voltage connection window of disconnected electric vehicle battery packs; and connecting one or more of the disconnected electric vehicle battery packs that are within the programmed voltage connection window of the disconnected electric vehicle battery packs to each other in one or more series strings.

In accordance with an exemplary embodiment, the method can further include connecting the electric vehicle battery packs in series/parallel within the programmed voltage connection window by calculating a histogram of the electric vehicle battery packs with binning determined by the programmed voltage window; and limiting charge or discharge current of the electric vehicle battery packs connected in series to a maximum programmed pack current. In addition, the method further includes disconnecting charged electric vehicle battery packs at an end of a charge cycle from the electric vehicle battery packs with lower voltages connected in series, the end of the charge cycle being based on one or more of a state of charge (SOC) or a cell voltage of the charged electric vehicle battery packs; and disconnecting discharged electric vehicle battery packs at an end of a discharge cycle from the electric vehicle battery packs with

14

higher voltages connected in series, the end of the discharge cycle being based on a state of charge (SOC) or a cell voltage of the discharged electric vehicle battery packs.

FIG. **10** illustrates a representative computer system **1000** in which embodiments of the present disclosure, or portions thereof, may be implemented as computer-readable code executed on hardware. For example, the BPC **140**, the SCADA **501**, and the power system manager **503** may be implemented in whole or in part by a computer system **1000** using hardware, software executed on hardware, firmware, non-transitory computer readable media having instructions stored thereon, or a combination thereof and may be implemented in one or more computer systems or other processing systems. Hardware, software executed on hardware, or any combination thereof may embody modules and components used to implement the methods and steps of the presently described method and system.

If programmable logic is used, such logic may execute on a commercially available processing platform configured by executable software code to become a specific purpose computer or a special purpose device (for example, programmable logic array, application-specific integrated circuit, etc.). A person having ordinary skill in the art may appreciate that embodiments of the disclosed subject matter can be practiced with various computer system configurations, including multi-core multiprocessor systems, minicomputers, mainframe computers, computers linked or clustered with distributed functions, as well as pervasive or miniature computers that may be embedded into virtually any device. For instance, at least one processor device and a memory may be used to implement the above described embodiments.

A processor unit or device as discussed herein may be a single processor, a plurality of processors, or combinations thereof. Processor devices may have one or more processor “cores.” The terms “computer program medium,” “non-transitory computer readable medium,” and “computer usable medium” as discussed herein are used to generally refer to tangible media such as a removable storage unit **1018**, a removable storage unit **1022**, and a hard disk installed in hard disk drive **1012**.

Various embodiments of the present disclosure are described in terms of this representative computer system **1000**. After reading this description, it will become apparent to a person skilled in the relevant art how to implement the present disclosure using other computer systems and/or computer architectures. Although operations may be described as a sequential process, some of the operations may in fact be performed in parallel, concurrently, and/or in a distributed environment, and with program code stored locally or remotely for access by single or multi-processor machines. In addition, in some embodiments the order of operations may be rearranged without departing from the spirit of the disclosed subject matter.

A processor device **1004** may be processor device specifically configured to perform the functions discussed herein. The processor device **1004** may be connected to a communications infrastructure **1006**, such as a bus, message queue, network, multi-core message-passing scheme, etc. The network may be any network suitable for performing the functions as disclosed herein and may include a local area network (“LAN”), a wide area network (“WAN”), a wireless network (e.g., “Wi-Fi”), a mobile communication network, a satellite network, the Internet, fiber optic, coaxial cable, infrared, radio frequency (“RF”), or any combination thereof. Other suitable network types and configurations will be apparent to persons having skill in the relevant art. The

15

computer system **1000** may also include a main memory **1008** (e.g., random access memory, read-only memory, etc.), and may also include a secondary memory **1010**. The secondary memory **1010** may include the hard disk drive **1012** and a removable storage drive **1014**, such as a floppy disk drive, a magnetic tape drive, an optical disk drive, a flash memory, etc.

The removable storage drive **1014** may read from and/or write to the removable storage unit **1018** in a well-known manner. The removable storage unit **1018** may include a removable storage media that may be read by and written to by the removable storage drive **1014**. For example, if the removable storage drive **1014** is a floppy disk drive or universal serial bus port, the removable storage unit **1018** may be a floppy disk or portable flash drive, respectively. In one embodiment, the removable storage unit **1018** may be non-transitory computer readable recording media.

In some embodiments, the secondary memory **1010** may include alternative means for allowing computer programs or other instructions to be loaded into the computer system **1000**, for example, the removable storage unit **1022** and an interface **1020**. Examples of such means may include a program cartridge and cartridge interface (e.g., as found in video game systems), a removable memory chip (e.g., EEPROM, PROM, etc.) and associated socket, and other removable storage units **1022** and interfaces **1020** as will be apparent to persons having skill in the relevant art.

Data stored in the computer system **1000** (e.g., in the main memory **1008** and/or the secondary memory **1010**) may be stored on any type of suitable computer readable media, such as optical storage (e.g., a compact disc, digital versatile disc, Blu-ray disc, etc.) or magnetic storage (e.g., a hard disk drive). The data may be configured in any type of suitable database configuration, such as a relational database, a structured query language (SQL) database, a distributed database, an object database, etc. Suitable configurations and storage types will be apparent to persons having skill in the relevant art.

The computer system **1000** may also include a communications interface **1024**. The communications interface **1024** may be configured to allow software and data to be transferred between the computer system **1000** and external devices. Exemplary communications interfaces **1024** may include a modem, a network interface (e.g., an Ethernet card), a communications port, a PCMCIA slot and card, etc. Software and data transferred via the communications interface **1024** may be in the form of signals, which may be electronic, electromagnetic, optical, or other signals as will be apparent to persons having skill in the relevant art. The signals may travel via a communications path **1026**, which may be configured to carry the signals and may be implemented using wire, cable, fiber optics, a phone line, a cellular phone link, a radio frequency link, etc.

The computer system **1000** may further include a display interface **1002**. The display interface **1002** may be configured to allow data to be transferred between the computer system **1000** and external display **1030**. Exemplary display interfaces **1002** may include high-definition multimedia interface (HDMI), digital visual interface (DVI), video graphics array (VGA), etc. The display **1030** may be any suitable type of display for displaying data transmitted via the display interface **1002** of the computer system **1000**, including a cathode ray tube (CRT) display, liquid crystal display (LCD), light-emitting diode (LED) display, capacitive touch display, thin-film transistor (TFT) display, etc.

Computer program medium and computer usable medium may refer to memories, such as the main memory **1008** and

16

secondary memory **1010**, which may be memory semiconductors (e.g., DRAMs, etc.). These computer program products may be means for providing software to the computer system **1000**. Computer programs (e.g., computer control logic) may be stored in the main memory **1008** and/or the secondary memory **1010**. Computer programs may also be received via the communications interface **1024**. Such computer programs, when executed, may enable computer system **1000** to implement the present methods as discussed herein. In particular, the computer programs, when executed, may enable processor device **1004** to implement the methods illustrated by FIGS. 1-9, as discussed herein. Accordingly, such computer programs may represent controllers of the computer system **1000**. Where the present disclosure is implemented using software executed on hardware, the software may be stored in a computer program product and loaded into the computer system **1000** using the removable storage drive **1014**, interface **1020**, and hard disk drive **1012**, or communications interface **1024**.

The processor device **1004** may comprise one or more modules or engines configured to perform the functions of the computer system **1000**. Each of the modules or engines may be implemented using hardware and, in some instances, may also utilize software executed on hardware, such as corresponding to program code and/or programs stored in the main memory **1008** or secondary memory **1010**. In such instances, program code may be compiled by the processor device **1004** (e.g., by a compiling module or engine) prior to execution by the hardware of the computer system **1000**. For example, the program code may be source code written in a programming language that is translated into a lower level language, such as assembly language or machine code, for execution by the processor device **1004** and/or any additional hardware components of the computer system **1000**. The process of compiling may include the use of lexical analysis, preprocessing, parsing, semantic analysis, syntax-directed translation, code generation, code optimization, and any other techniques that may be suitable for translation of program code into a lower level language suitable for controlling the computer system **1000** to perform the functions disclosed herein. It will be apparent to persons having skill in the relevant art that such processes result in the computer system **1000** being a specially configured computer system **1000** uniquely programmed to perform the functions discussed above.

Techniques consistent with the present disclosure provide, among other features, an energy storage system and method employing second-life electric vehicle batteries. While various exemplary embodiments of the disclosed system and method have been described above it should be understood that they have been presented for purposes of example only, not limitations. It is not exhaustive and does not limit the disclosure to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practicing of the disclosure, without departing from the breadth or scope.

What is claimed is:

1. An integrated battery energy storage system, the integrated battery energy system comprising:
 - a plurality of electric vehicle battery packs; and
 - a computer system, the computer system including a processor configured to:
 - couple the plurality of electric vehicle battery packs in a series/parallel arrangement, the series/parallel arrangement including a plurality of series strings of electric vehicle battery packs, each of the plurality of series strings of electric vehicle battery packs

17

- includes at least two of the plurality of electric vehicle battery packs coupled in series, and wherein the plurality of series strings of electric vehicle battery packs are connected in parallel; and wherein the coupling of the plurality of electric vehicle battery packs includes one or more of connecting electric vehicle battery packs with lower voltages in series, connecting electric vehicle battery packs with higher voltages in series, connecting electric vehicle battery packs with majority voltages in series, and connecting electric vehicle battery packs within a programmed voltage connection window in parallel.
2. The integrated battery energy storage system according to claim 1, wherein the processor is configured to: connect the electric vehicle battery packs with lower voltages in series as at least a lower voltage series string; and charge the lower voltage series string to a maximum programmed pack current.
3. The integrated battery energy storage system according to claim 2, wherein one or more of the electric vehicle battery packs with the lower voltages is faulted, the processor is configured to: connect one or more electric vehicle battery packs with a next lowest voltage to the lower voltage series string in place of the one or more the electric vehicle battery packs with the lower voltages that is faulted.
4. The integrated battery energy storage system according to claim 2, wherein the processor is configured to: charge one or more of the electric vehicle battery packs connected in the lower voltage series string to a maximum programmed pack current of one or more disconnected electric vehicle battery packs; connect in parallel one or more of the electric vehicle battery packs within a programmed voltage connection window of disconnected electric vehicle battery packs; and connect one or more of the disconnected electric vehicle battery packs that are within the programmed voltage connection window of the disconnected electric vehicle battery packs to each other in one or more series strings.
5. The integrated battery energy storage system according to claim 1, wherein the processor is configured to: connect the electric vehicle battery packs with higher voltages in series as a higher voltage series string; and discharge the higher voltage series string to a maximum programmed pack current.
6. The integrated battery energy storage system according to claim 5, wherein one or more of the electric vehicle battery packs with the higher voltages is faulted, the processor is configured to: connect one or more electric vehicle battery packs with a next highest voltage to the higher voltage series string in place of the one or more the electric vehicle battery packs with the higher voltages that is faulted.
7. The integrated battery energy storage system according to claim 5, wherein the processor is configured to: discharge one or more of the electric vehicle battery packs connected in the higher voltage series string to a maximum programmed pack current of one or more disconnected electric vehicle battery packs; connect in parallel one or more of the electric vehicle battery packs within a programmed voltage connection window of disconnected electric vehicle battery packs; and

18

- connect one or more of the disconnected electric vehicle battery packs that are within the programmed voltage connection window of the disconnected electric vehicle battery packs to each other in one or more series strings.
8. The integrated battery energy storage system according to claim 1, wherein the processor is configured to: connect the electric vehicle battery packs in series/parallel within the programmed voltage connection window by calculating a histogram of the electric vehicle battery packs with binning determined by the programmed voltage window; and limit charge or discharge current of the electric vehicle battery packs connected in series to a maximum programmed pack current.
9. The integrated battery energy storage system according to claim 1, wherein the processor is configured to: disconnect charged electric vehicle battery packs at an end of a charge cycle from the electric vehicle battery packs with lower voltages connected in series, the end of the charge cycle being based on one or more of a state of charge (SOC) or a cell voltage of the charged electric vehicle battery packs; and disconnect discharged electric vehicle battery packs at an end of a discharge cycle from the electric vehicle battery packs with higher voltages connected in series, the end of the discharge cycle being based on a state of charge (SOC) or a cell voltage of the discharged electric vehicle battery packs.
10. The integrated battery energy storage system according to claim 1, further comprising: an external cell balancer, the external cell balancer configured to maintain one or more intermediate rails of the plurality of electric vehicle battery packs coupled in the series/parallel arrangement at a same voltage; a circuit combiner comprising an overcurrent device in series with each of the plurality series strings; and a smart combiner comprising a plurality of DC-to-DC converters configured to balance two or more electric vehicle battery packs in a common series string by moving energy from an electric vehicle battery pack to another electrical vehicle battery pack in a common series string or by moving energy to an electric vehicle battery pack from another electrical vehicle battery pack in a common series string.
11. The integrated battery energy storage system according to claim 1, further comprising: a battery pack controller configured to communicate with and control a battery management system within each of the plurality of electric vehicle battery packs; and a bidirectional power converter configured to convert DC electric vehicle battery pack power to AC electrical grid power or to convert AC electrical grid power to charge DC electric vehicle battery packs.
12. A method for integrating electric vehicle battery packs into an integrated battery energy storage system, the method comprising: coupling a plurality of electric vehicle battery packs in a series/parallel arrangement, the series/parallel arrangement including a plurality of series strings of electric vehicle battery packs, each of the plurality of series strings of electric vehicle battery packs includes at least two of the plurality of electric vehicle battery packs coupled in series, and wherein the plurality of series strings of electric vehicle battery packs are connected in parallel; and

19

wherein the coupling of the plurality of electric vehicle battery packs comprises one or more of connecting electric vehicle battery packs with lower voltages in series, connecting electric vehicle battery packs with higher voltages in series, connecting electric vehicle battery packs with majority voltages in series, and connecting electric vehicle battery packs within a programmed voltage connection window in parallel.

13. The method according to claim 12, further comprising:

connecting the electric vehicle battery packs with lower voltages in series as at least a lower voltage series string; and
charging the lower voltage series string to a maximum programmed pack current.

14. The method according to claim 13, wherein one or more of the electric vehicle battery packs with the lower voltages is faulted, further comprising:

connecting one or more electric vehicle battery packs with a next lowest voltage to the lower voltage series string in place of the one or more the electric vehicle battery packs with the lower voltages that is faulted.

15. The method according to claim 13, further comprising:

charging one or more of the electric vehicle battery packs connected in the lower voltage series string to a maximum programmed pack current of one or more disconnected electric vehicle battery packs;

connecting in parallel one or more of the electric vehicle battery packs within a programmed voltage connection window of disconnected electric vehicle battery packs; and

connecting one or more of the disconnected electric vehicle battery packs that are within the programmed voltage connection window of the disconnected electric vehicle battery packs to each other in one or more series strings.

16. The method according to claim 12, further comprising:

connecting the electric vehicle battery packs with higher voltages in series as a higher voltage series string; and
discharging the higher voltage series string to a maximum programmed pack current.

17. The method according to claim 16, wherein one or more of the electric vehicle battery packs with the higher voltages is faulted, further comprising:

20

connecting one or more electric vehicle battery packs with a next highest voltage to the higher voltage series string in place of the one or more the electric vehicle battery packs with the higher voltages that is faulted.

18. The method according to claim 16, further comprising:

discharging one or more of the electric vehicle battery packs connected in the higher voltage series string to a maximum programmed pack current of one or more disconnected electric vehicle battery packs;

connecting in parallel one or more of the electric vehicle battery packs within a programmed voltage connection window of disconnected electric vehicle battery packs; and

connecting one or more of the disconnected electric vehicle battery packs that are within the programmed voltage connection window of the disconnected electric vehicle battery packs to each other in one or more series strings.

19. The method according to claim 12, further comprising:

connecting the electric vehicle battery packs in series/parallel within the programmed voltage connection window by calculating a histogram of the electric vehicle battery packs with binning determined by the programmed voltage window; and

limiting charge or discharge current of the electric vehicle battery packs connected in series to a maximum programmed pack current.

20. The method according to claim 12, further comprising:

disconnecting charged electric vehicle battery packs at an end of a charge cycle from the electric vehicle battery packs with lower voltages connected in series, the end of the charge cycle being based on one or more of a state of charge (SOC) or a cell voltage of the charged electric vehicle battery packs; and

disconnecting discharged electric vehicle battery packs at an end of a discharge cycle from the electric vehicle battery packs with higher voltages connected in series, the end of the discharge cycle being based on a state of charge (SOC) or a cell voltage of the discharged electric vehicle battery packs.

* * * * *